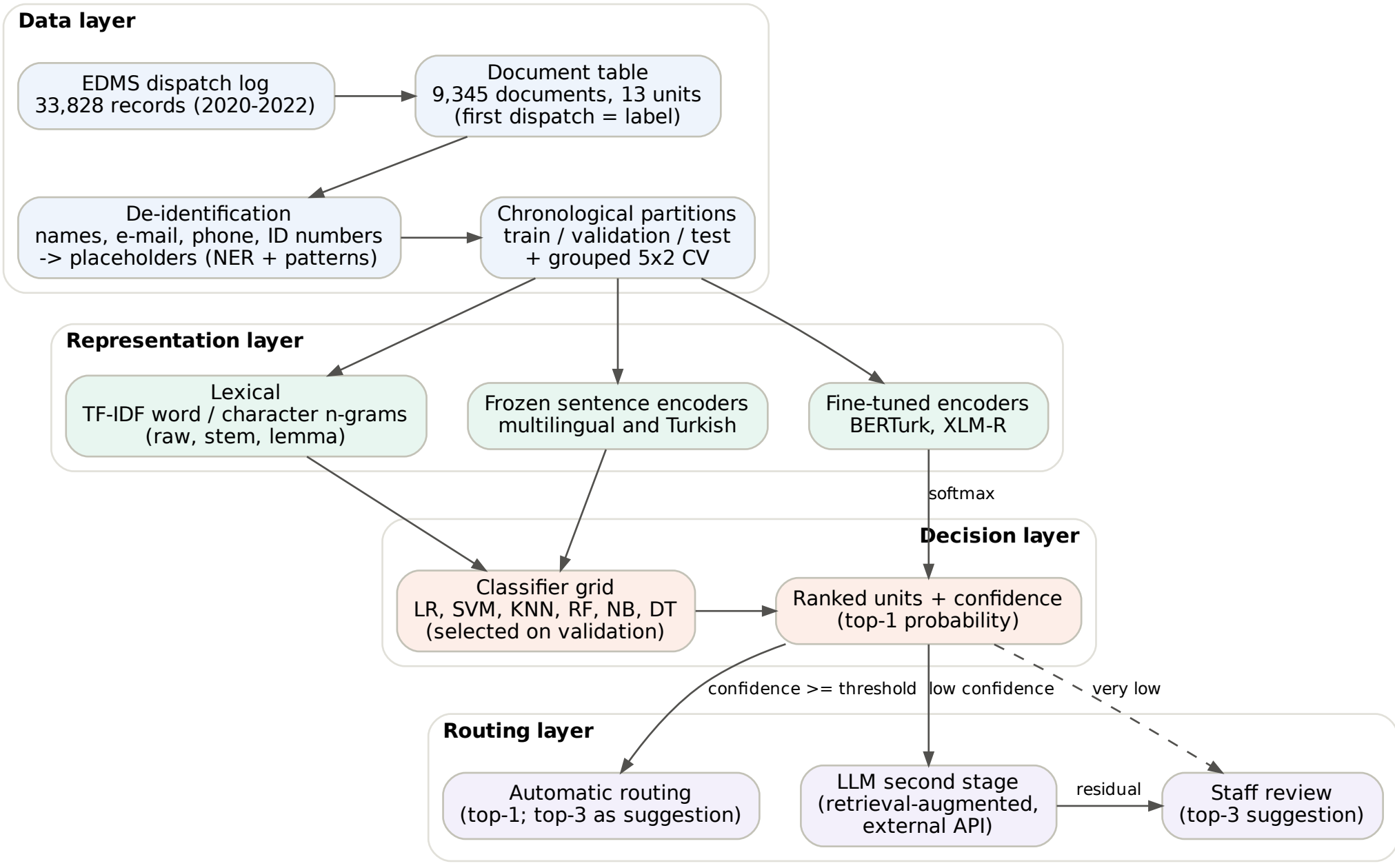
Data layer
EDMS dispatch log
33,828 records (2020-2022)
Document table
9,345 documents, 13 units
(first dispatch = label)
De-identification
names, e-mail, phone, ID numbers
-> placeholders (NER + patterns)
Chronological partitions
train / validation / test
+ grouped 5x2 CV
Representation layer
Lexical
TF-IDF word / character n-grams
(raw, stem, lemma)
Frozen sentence encoders
multilingual and Turkish
Fine-tuned encoders
BERTurk, XLM-R
softmax
Decision layer
Classifier grid
LR, SVM, KNN, RF, NB, DT
(selected on validation)
Ranked units + confidence
(top-1 probability)
confidence >= threshold
low confidence
very low
Routing layer
Automatic routing
(top-1; top-3 as suggestion)
LLM second stage
(retrieval-augmented,
external API)
residual
Staff review
(top-3 suggestion)